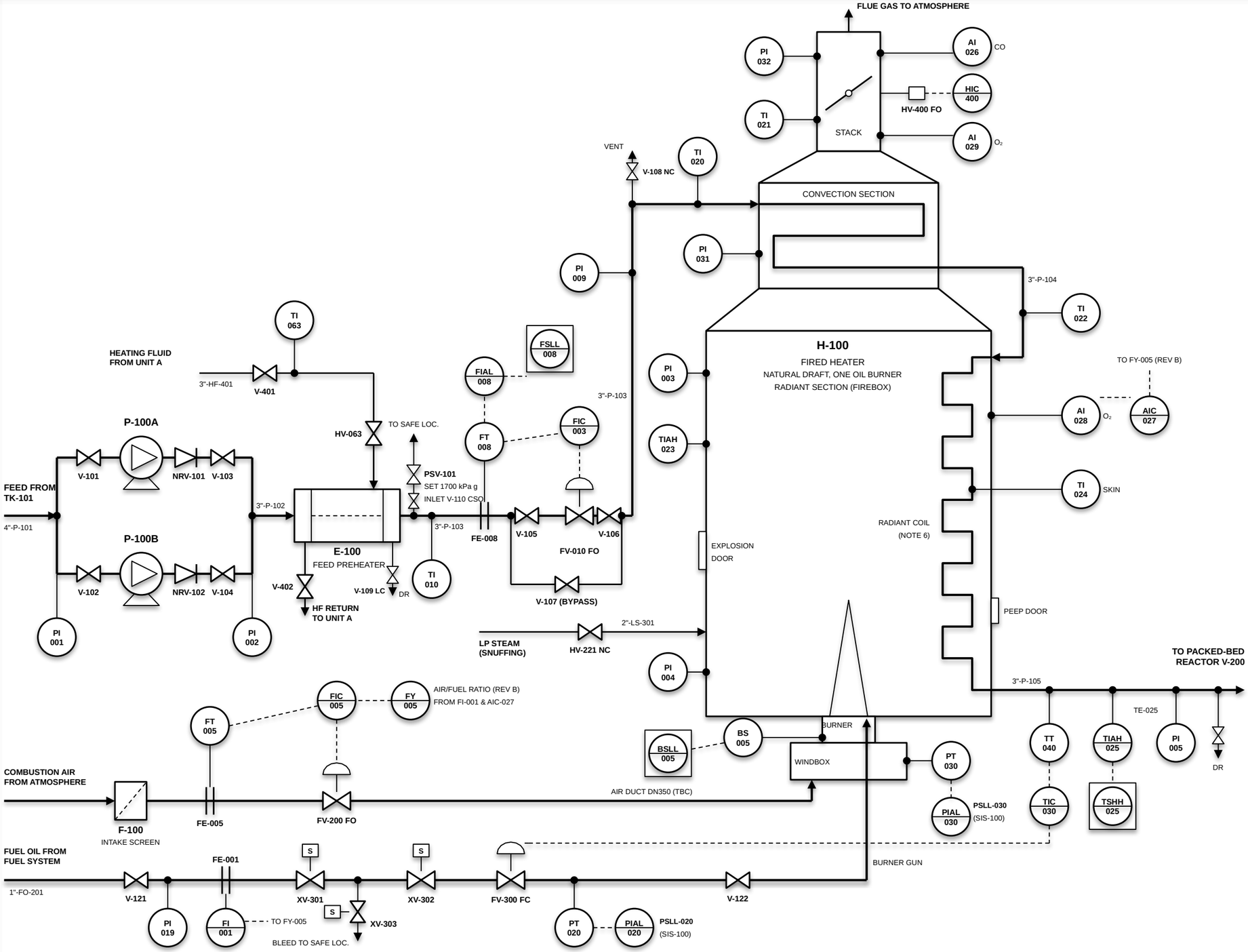


LEGEND

- PROCESS LINE
- UTILITY LINE
- INSTRUMENT SIGNAL
- MANUAL VALVE (GATE)
- CONTROL VALVE, FO / FC = FAIL OPEN / CLOSED
- SIS SHUT-OFF VALVE, SOLENOID
- CHECK VALVE
- FLOW ELEMENT (ORIFICE)
- RELIEF VALVE (PSV)
- FIELD-MOUNTED INSTRUMENT (PI 001)
- DCS FUNCTION (OPERATOR CONSOLE) (FIC 003)
- SIS-100 / BMS FUNCTION (TSHH 025)

SERVICE CODES: P PROCESS HYDROCARBON · FO FUEL OIL
HF HEATING FLUID · LS LP STEAM · DR DRAIN
LINE NO. = SIZE - SERVICE - NUMBER

INTERLOCK SCHEDULE · SIS-100 / BMS

TRIP	CAUSE	SETTING	DELAY
FSLL-008	LOW FEED FLOW	< 3.4 L/s	5 s
TSHH-025	HIGH OUTLET TEMP.	> 375 °C	5 s
PSLL-020	LOW FUEL P AT GUN*	< 40 kPa	3 s
PSLL-030	LOSS OF WINDBOX DRAFT*	< 0.5 mmH_2O	3 s
BSLL-005	FLAME FAILURE	NO FLAME	2 s
HS-001	ESD PUSHBUTTON	MANUAL	0 s

* ARMED WHEN FIRING.
ANY TRIP: CLOSE XV-301 & XV-302, OPEN XV-303 BLEED,
CLOSE FV-300, OPEN FV-200 & HV-400 (POST-PURGE).
RESET ONLY FROM BMS AFTER 300 s PURGE; LIGHT-OFF
AT LOW FIRE (TIC-030 MAN 10-20 %), 10 s TRIAL.

GENERAL NOTES:

1. DESIGN BASIS: NATURAL DRAFT FIRED HEATER VER 1 OTS MODEL, DESIGN CASE. PRESSURES kPa g, DRAFT mmH_2O.
   FEED 6.8 L/s = 19 584 kg/h (0.80 kg/L): 40 → 180 (E-100) → 233 (CROSSOVER) → 350 °C; PUMP DISCH. ≈ 1030, OUTLET ≈ 545.
   FUEL 243 kg/h, LHV 42 MJ/kg: FIRED 2.84 MW, ABSORBED 2.31 MW (81.5 %). AIR 3 892 kg/h (11 % EXCESS).
   FLUE GAS 4 380 kg/h AT 400 °C (INCL. ≈ 245 kg/h TRAMP AIR). DRAFT: ARCH -2.5, FLOOR -6.0, WINDBOX 5.25.
2. LINE SIZES PRELIMINARY FROM VELOCITY (LIQUID 1.5-2.5 m/s, AIR DUCT 10 m/s) - TBC BY HYDRAULIC CALCULATION.
3. PRESSURE INSTRUMENTS C/W ROOT AND TEST VALVES (TYP., NOT SHOWN). PUMP CASING VENTS/DRAINS TYP., NOT SHOWN.
4. XV-301 / XV-302 / XV-303 FORM A DOUBLE BLOCK AND BLEED, OPERATED BY SIS-100 (BMS); SIGNALS NOT SHOWN.
5. FV-010 FAILS OPEN SO THE COIL KEEPS FLOW ON AIR OR SIGNAL FAILURE; FV-300 FAILS CLOSED; FV-200 & HV-400 FAIL OPEN.
6. RADIANT AND CONVECTION COILS SHOWN SCHEMATICALLY; TUBE COUNT AND PASSES PER DESIGN TAB (SINGLE PASS).
7. FT-008 SERVES FIC-003, FIAL-008 AND FSLL-008 (HAZOP GAP: ADD FT-008B FOR THE TRIP - HOLD).

NOTES (CONT.):

8. TIC-030 MOVES FUEL ONLY. REV A: FIC-005 ON OWN SET POINT. REV B OPTION: FY-005 AIR/FUEL RATIO ON FI-001 + AIC-027 O_2 TRIM.
9. PSV-101 PROTECTS E-100 FEED SIDE; SET 1700 kPa g, F ORIFICE, GOVERNED BY FIRE (RELIEF LOAD BASIS NDFH-V1-PE-RLB-001).
10. PI-019 = FUEL HEADER (≈ 1940 kPa g). PT-020 = GUN PRESSURE (≈ 600 kPa g), DOWNSTREAM OF FV-300.
11. BS-005 UV FLAME SCANNER ON THE BURNER; LIGHT-OFF AND PURGE PER BMS SEQUENCE (LEGEND PANEL).
12. HV-221 SNUFFING STEAM, NORMALLY CLOSED; OPEN ONLY FOR A FIREBOX FIRE. EXPLOSION DOOR ON RADIANT WALL.
13. V-108 HIGH-POINT VENT AT COIL INLET (NC); V-109 E-100 CHANNEL DRAIN (LC); V-110 PSV-101 INLET (CSO). REV E: HAND VALVES TAGGED FOR THE 3-D PLANT.
14. ALARM SET POINTS: SEE WORKBOOK (ALARM TABLE); TRIP SET POINTS: INTERLOCK SCHEDULE.

REV	DESCRIPTION	DATE
B	RATIO FY-005 + O_2 TRIM AIC-027 (OPTION)	--
C	OTS SCREEN DRAWING (CURRENT, USED IN DECKS)	--
D	ALTERNATIVE VERSION, EPC STANDARD	29-SEP-26
E	HAND VALVES V-108 TO V-110, V-401, V-402 TAGGED	25-OCT-26

NATURAL DRAFT FIRED HEATER VER 1 · OTS

P&ID — H-100 · ALTERNATIVE VERSION

DRAWN: APINAN SOOTTITANTAWAT	CHK'D: --	APP'D: --	DATE: 25-OCT-26
DWG NO. NDFH-V1-PID-001	REV E	SCALE N.T.S.	SHEET 1 OF 1